

Introduction To Development With Azure Functions

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About Me



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Overview

According to the Microsoft documentation:

Azure Functions is a serverless solution that allows you to write less code, maintain less infrastructure, and save on costs. Instead of worrying about deploying and maintaining servers, the cloud infrastructure provides all the up-to-date resources needed to keep your applications running.

You focus on the code that matters most to you, in the most productive language for you, and Azure Functions handles the rest.

<https://learn.microsoft.com/en-us/azure/azure-functions/functions-overview>

The above may or may not be true depending on the choice you make for hosting.

Hosting Plans

- Consumption (Serverless)
 - Only pay for execution time (i.e. when functions are used)
 - Can be delays due to cold worker starts
- Flex Consumption (Serverless)
 - Similar to plan above
 - Optionally configure one or more pre-warmed instances
 - Pay for execution time plus cost of pre-warmed instances
- Premium
 - Requires an Azure App Service
 - One or more pre-warmed instances
 - Additional instances added as needed
 - Can deploy multiple Function Apps to same App Service

Hosting Plans 2

- App Service (Dedicated)
 - Requires an Azure App Service
 - You manage hosting
- Container Apps
 - Requires an Azure App Service
 - Function Apps are containerized
 - You manage hosting

Supported Languages

Depending on the hosting plan, you can use the following languages:

- C#
- TypeScript, JavaScript
- PowerShell
- Java
- Python

.NET Runtimes

Azure Functions currently supports two versions of the runtime host

- v1
 - Supported only for C# apps that must use .NET Framework
 - Support will end in September 2026
- v4
 - Recommended runtime version for functions in all languages

.NET: In-process vs isolated

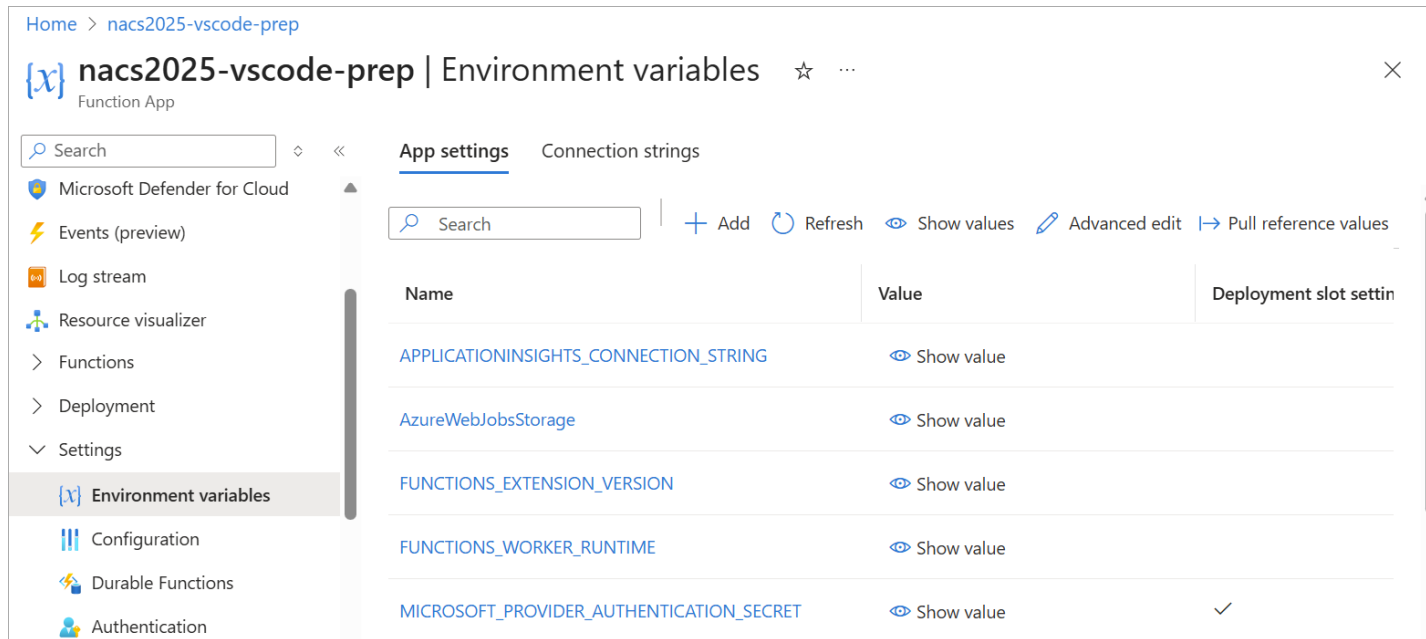
- In-process
 - Function code runs in the same process as the Functions host process
 - Support for in-process model ends in November 2026
- Isolated
 - Your function code runs in a separate .NET worker process

Triggers and Bindings

- Triggers
 - Causes a function to run
 - Each function must have exactly one trigger
- Bindings
 - Declaratively connecting another resource to the function
 - May be input, output or both
 - Data from bindings is provided to the function as parameters

App Settings (Environment Variables)

- Running locally
 - Configured in local.settings.json file in project
- Running from Azure host
 - Configured in Azure portal



The screenshot displays the Azure portal interface for a Function App named 'nacs2025-vscode-prep'. The 'Environment variables' tab is selected, showing a list of application settings. The left sidebar contains navigation options like 'Microsoft Defender for Cloud', 'Events (preview)', 'Log stream', 'Resource visualizer', 'Functions', 'Deployment', 'Settings', 'Environment variables', 'Configuration', 'Durable Functions', and 'Authentication'. The main content area has a search bar and buttons for '+ Add', 'Refresh', 'Show values', 'Advanced edit', and 'Pull reference values'. Below these is a table with columns for Name, Value, and Deployment slot settings.

Name	Value	Deployment slot settings
APPLICATIONINSIGHTS_CONNECTION_STRING	Show value	
AzureWebJobsStorage	Show value	
FUNCTIONS_EXTENSION_VERSION	Show value	
FUNCTIONS_WORKER_RUNTIME	Show value	
MICROSOFT_PROVIDER_AUTHENTICATION_SECRET	Show value	✓

Disable Function

Function overview page

Home > nacs2025-vscode-prep >

>HelloWorld | Code + Test

nacs2025-vscode-prep

Code + Test

Integration

Function Keys

Invocations

Logs

Metrics

Save

Discard

Refresh

Test/Run

Get function URL

Disable

Delete

Upload

Resource JSON

Send us your feedback

Editing .NET isolated Function Apps is not supported in the Azure portal. Use your local development environment to edit this Function App.

nacs2025-vscode-prep / functions.metadata

1

2

3

4

5

6

7

8

[

{

...."name": "HelloWorld",

...."scriptFile": "AzureFunctionCodeCS.dll",

...."entryPoint": "AzureFunctionCodeCS.HelloWorld.Run",

...."language": "dotnet-isolated",

...."properties": {

...."IsCodeless": false

}

}

Logs

App Insights Logs

Log Level

Stop

Copy

Clear

Maximize

Send us your feedback

Connected! You are now viewing logs of Function runs in the current Code + Test panel. To see all the logs for this Function, please go to 'Logs' from the Function menu.

Environment variable

Home > nacs2025-vscode-prep

{x} nacs2025-vscode-prep | Environment variables

Function App

Search

disabled

+ Add

Refresh

App settings

Connection strings

Functions

Deployment

Settings

Environment variables

Configuration

Durable Functions

Authentication

Azure OpenAI

Name

Value

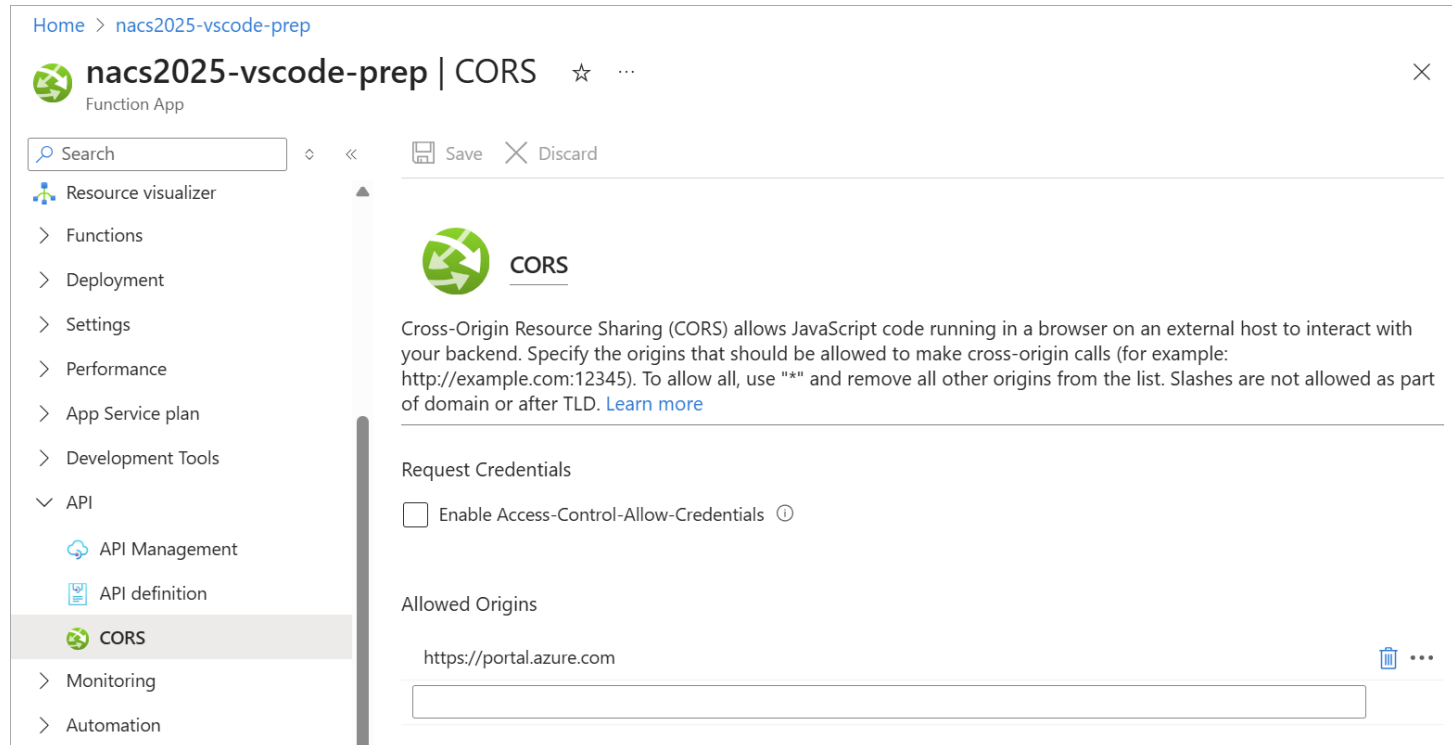
AzureWebJobs.HelloWorld2.Disabled

1

CORS

Cross-Origin Resource Sharing (CORS) allows JavaScript code running in a browser on an external host to interact with your backend.

- Configured in Azure portal



Durable Functions

An extension of Azure Functions that lets you write stateful functions even in a serverless compute environment.

- Define stateful workflows by writing orchestrator functions
- Define stateful entities by writing entity functions
- Extension manages state, checkpoints, and restarts for you

Resources

Azure Functions Documentation

<https://learn.microsoft.com/en-us/azure/azure-functions/>

Azure Functions Core Tools

<https://learn.microsoft.com/en-us/azure/azure-functions/functions-run-local>

Azure CLI

<https://learn.microsoft.com/en-us/cli/azure/>